

Tutorial 38

Problem set 38: Miller's clawback, the multiplier, the loop

Advanced Corporate Finance (EBM232A05)

University of Groningen

- **The spine: Problem set 38**, all five problems — task, then the decisive steps at the board, then the numbers checked against these slides.
- Up front, **ten minutes of project craft**: what a regression table's cells are, and the three places papers stay silent — you start pulling data this week, and these are the choices waiting for you.

Project status: scoping memos came in last night (22:00). Written feedback and the proceed/no-proceed verdict come back **next week** — assessment moment M1. No project work due today.

Ten minutes of project craft

Your target cells are regression output

Open your assigned paper at its headline table. Whatever the topic, the object is

$$y_{it} = \beta x_{it} + \gamma' z_{it} + \alpha_i + \delta_t + \varepsilon_{it}, \quad \text{firms } i, \text{ years } t.$$

- Your **target cells** are estimates of β , their standard errors, N , and R^2 — nothing else.
- To land in the tolerance band you must make the authors' choices: **sample, variable construction, fixed effects, clustering**. Each is a row of the variable-definition table your scoping memo drafted.

You know the econometrics from your prerequisite course. What is new is the **craft**: which conventions empirical corporate finance applies without saying so.

The three silent defaults (= your D3 catalogue)

Where the paper is silent, the AI guesses — its favourite wrong guesses:

- **Clustering.** CF panels cluster by **firm**; AI's default is often plain *robust*. Symptom: coefficient in band, **SE half the paper's**. Both are target cells.
- **Winsorization.** Usually 1%/99% — but on **which sample**, before or after filters? Moves coefficients by percents.
- **Sample filters.** Financials (SIC 6000–6999) and utilities (4900–4999) are dropped by convention — **does your paper say so?**

Each is a ready-made entry for the **underspecification catalogue** (D3): AI default vs. your choice vs. how the target cell moves.

Problem set 38, discussed

One euro of operating income, two routes to the investor:

	as interest	as equity income
corporate layer	— (deductible)	τ_c
personal layer	τ_d	τ_e
investor keeps	$1 - \tau_d$	$(1 - \tau_c)(1 - \tau_e)$

At $\tau_c = 20\%$, $\tau_e = 10\%$:

- (a) derive G_L from the two survival rates;
- (b) compute G_L/D at $\tau_d = 25\%$ and $\tau_d = 40\%$ — sign both;
- (c) find τ_d^* , and argue why the market pushes the marginal investor there;
- (d) what does this make of the net shield τ in Week 37's thresholds?

$$G_L = \left[1 - \frac{(1 - \tau_c)(1 - \tau_e)}{1 - \tau_d} \right] D \quad \text{— the ratio of the two survival rates.}$$

With $(1 - \tau_c)(1 - \tau_e) = 0.72$:

- $\tau_d = 25\%$: $1 - 0.72/0.75 = +4\% \uparrow$ $\tau_d = 40\%$: $1 - 0.72/0.60 = -20\% \downarrow$
- Indifference: $1 - \tau_d^* = 0.72 \Rightarrow \tau_d^* = 28\%$. If $G_L > 0$, firms issue more debt and must place it in ever higher brackets — until the **marginal** investor sits exactly there (Miller, 1977).
- (d) The τ of Week 37's thresholds is the **net-of-clawback** shield: $+4\%$, not $\tau_c = 20\%$. A small τ makes τI cheap to give up — **that is why the flexibility motive wins so often**.

One friction: outside money is capped by collateral,

$$I = w + b, \quad b \leq \lambda w.$$

- (a) Combine budget and limit into one expression for I ; find the threshold w^* .
- (b) Compute dI/dw on each side. Why exactly $1 + \lambda$?
- (c) “Sensitivity is highest for the most constrained firms” — does the model say that?
- (d) What raises λ ? What is the $\lambda \rightarrow \infty$ limit?

Board first — the slide after this one has the answers.

$$I = \min\{I^*, (1 + \lambda)w\}, \quad w^* = \frac{I^*}{1 + \lambda}, \quad \frac{dI}{dw} = \begin{cases} 1 + \lambda & w < w^* \\ 0 & w > w^* \end{cases}$$

- The euro enters the budget itself (+1) *and* relaxes the limit by λ : net worth is **borrowing capacity**, not just money.
- (c) No: sensitivity is $1 + \lambda$ on the ray, 0 past w^* — **not monotone** in constrainedness. A noisy sort around w^* *predicts* Kaplan and Zingales (1997)'s finding: the objection is the model's own comparative static.
- (d) λ : tangibility, creditor rights, transparency. $\lambda \rightarrow \infty$: capacity unbounded — **MM separation**, the balance sheet drops out.

$w = 60$, $\lambda = 1.5$, $I^* = 180$. A recession destroys $\Delta w = 10$.

	Net worth w	Borrowing b	Investment I
Before	60	?	?
After the shock	50	?	?
Frictionless, after	50	?	?

Then: the cost of the shock per euro of net worth; the transfer that restores first-best — and what one euro of transfer buys *before* and *after* that point.

P3 — solved

	Net worth w	Borrowing b	Investment I
Before	60	90	150 (shortfall 30; $w^* = 72$)
After the shock	50	75	125 ↓
Frictionless, after	50	130 ↑	180

- The shock costs $\Delta I = 25 = 2.5 \times \Delta w$: the euro itself plus the €1.50 of borrowing it supported. The frictionless firm simply borrows **more** — exactly what the limit forbids.
- Transfer: while constrained, €1 buys **€2.50** of investment; first-best needs $w \geq 72$, i.e. a transfer of **22** — at which point the multiplier drops to **0**.

One number, three appearances: amplification, transfer productivity, slope of the ray. It is **one object**.

All firms cut together, so the collateral price moves. Per round:

$$\frac{\Delta q}{q} = \frac{\Delta I_{\text{prev}}}{500}, \quad \text{€48 of net worth is marked to market.}$$

- (a) Round 1: the shock cut investment by $\Delta I_0 = 25$. Price, net-worth loss, ΔI_1 ?
- (b) Round 2: ΔI_2 ?
- (c) **Name the pattern** before computing round 3 — then the total in closed form.
- (d) When does the sum fail to converge, and what are the two policy dials inside your condition?

P4 — solved

Round 1: $\Delta q/q = 5\%$, $\Delta w_1 = 2.4$, $\Delta I_1 = \mathbf{6}$. Round 2: $\Delta q/q = 1.2\%$, $\Delta w_2 = 0.576$, $\Delta I_2 = \mathbf{1.44}$.

$$\theta = (1 + \lambda) \cdot \frac{48}{500} = 0.24, \quad \Delta I_{\text{total}} = \frac{\Delta I_0}{1 - \theta} = \frac{25}{0.76} \approx \mathbf{32.9}.$$

- The price feedback adds a factor $1/(1 - \theta) \approx 1.32$ on top of the within-firm 2.5: total ≈ 3.3 per euro of destroyed net worth. Kiyotaki and Moore (1997), stripped to arithmetic. (θ , not φ — that letter is dilution, Week 41.)
- (d) Divergence at $\theta \geq 1$: **fire-sale spiral**. The condition contains the two dials regulation uses — cap the system's leverage (λ : capital and margin requirements), and deepen or backstop the asset market in a crisis. **Week 42 returns to the first dial.**

Three papers, one design lesson:

- (a) Why can the investment–cash-flow regression (Fazzari et al., 1988) not settle the question on its own?
- (b) What do Almeida et al. (2004) change — and why does a noisy q no longer matter?
- (c) Chaney et al. (2012): six cents of investment per euro of real-estate collateral. Which model object is that, and why does real estate isolate it?
- (d) Write the Week-38 ledger row.

- (a) q is a noisy proxy, and cash flow carries the news q missed — large coefficients with **no** friction at all.
- (b) A **sign difference** across groups: constrained firms save $\approx 5\text{--}6$ cents of each cash-flow euro as cash, unconstrained ≈ 0 ; mismeasured q inflates both alike. **Designs whose predictions differ in sign survive mismeasurement.**
- (c) The constrained branch's slope — λ : local real-estate prices move the *buildings*, not the *projects*.

The design lesson in (b) is project-relevant: when your reproduction misses, ask whether the miss is a **level** problem (vintage, filters) or a **sign/structure** problem (specification) — they have different root causes.

The Week-38 ledger row

Wk	Friction		Effect		Evidence
38	Costly finance pledgeability)	external (limited	investment ↓, plified $\times(1+\lambda)$; hoarding ↑	am- cash	cash sensitivity of cash (Almeida et al., 2004); collat- eral channel (Chaney et al., 2012)

Same row as the lecture's close — the problem set's contribution is that you can now **derive** the amplification instead of quoting it.

Before you leave

- **Project:** nothing due this week. M1 feedback + proceed verdicts at next week's tutorial. Meanwhile: WRDS pulls belong to **you**, not to the AI — record query specs and pull dates as you go.
- **Problem set 38:** the solution key is on Brightspace after today. The two Explorers (Miller clawback; net-worth multiplier) are the self-check: sliders show the direction, your derivation shows why.
- **Next week** (lecture): the manager wedge — why pay packages look the way they do. Tutorial: M1 verdicts + Problem set 39.

- Almeida, H., M. Campello, and M. S. Weisbach (2004). **“The Cash Flow Sensitivity of Cash”**. In: *The Journal of Finance* 59.4, pp. 1777–1804.
- Chaney, T., D. Sraer, and D. Thesmar (2012). **“The Collateral Channel: How Real Estate Shocks Affect Corporate Investment”**. In: *The American Economic Review* 102.6, pp. 2381–2409.
- Fazzari, S. M., R. G. Hubbard, and B. C. Petersen (1988). **“Financing Constraints and Corporate Investment”**. In: *Brookings Papers on Economic Activity* 1988.1, pp. 141–206.
- Kaplan, S. N. and L. Zingales (1997). **“Do Investment–Cash Flow Sensitivities Provide Useful Measures of Financing Constraints?”** In: *The Quarterly Journal of Economics* 112.1, pp. 169–215.

- Kiyotaki, N. and J. Moore (1997). **“Credit Cycles”**. In: *Journal of Political Economy* 105.2, pp. 211–248.
- Miller, M. H. (1977). **“Debt and Taxes”**. In: *The Journal of Finance* 32.2, pp. 261–275.